Sukkur Institute of Business Administration University
Department of Computer Science

Software Quality Engineering
BS – V (Software Engineering)
Fall 2026

Instructor: Dr Khursheed Ali

LAB # 009

Submission Profile

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Semester: ‘5’

Section: ‘A’

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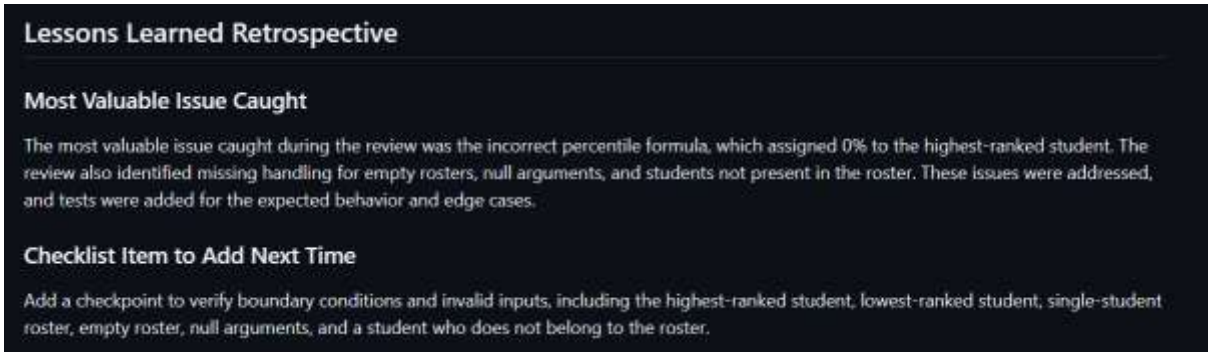
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GitHub repository URL

https://github.com/anasqadri-dev/software-quality-engineering

Code Review Checklist and Retrospective

https://github.com/anasqadri-dev/software-quality-engineering/blob/main/LABs/LAB%2309/sqe-gradebook/docs/code-review-checklist.md



Lessons Learned Retrospective

Most Valuable Issue Caught

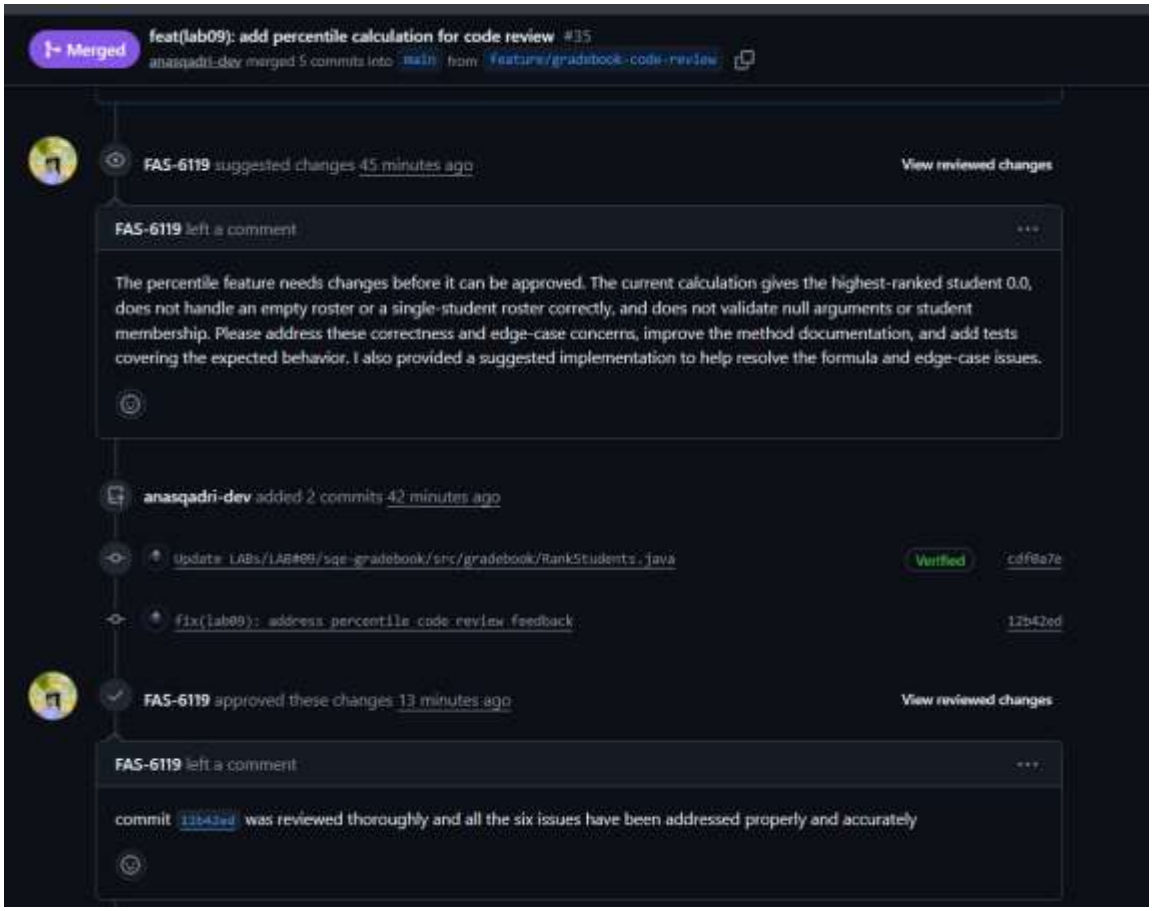
The most valuable issue caught during the review was the incorrect percentile formula, which assigned 0% to the highest-ranked student. The review also identified missing handling for empty rosters, null arguments, and students not present in the roster. These issues were addressed, and tests were added for the expected behavior and edge cases.

Checklist Item to Add Next Time

Add a checkpoint to verify boundary conditions and invalid inputs, including the highest-ranked student, lowest-ranked student, single-student roster, empty roster, null arguments, and a student who does not belong to the roster.

Merged GitHub Pull Request and Review Conversation

https://github.com/anasqadri-dev/software-quality-engineering/pull/35



Final percentile() Implementation and Edge-Case Testing

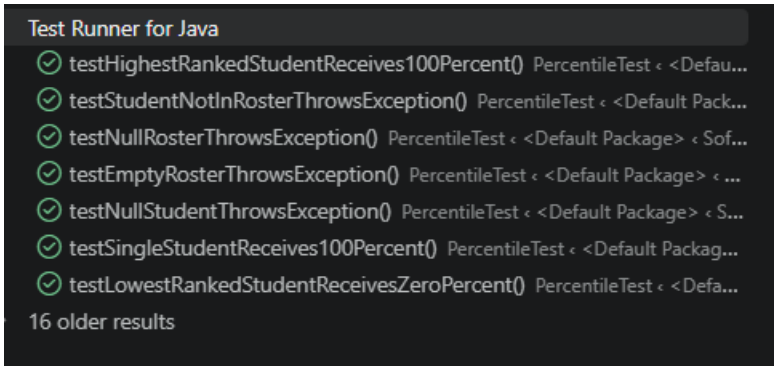
The complete final percentile() implementation from src/gradebook/RankStudents.java.

https://github.com/anasqadri-dev/software-quality-engineering/blob/main/LABs/LAB%2309/sqe-gradebook/src/gradebook/RankStudents.java

The edge-case tests from tests/PercentileTest.java.

https://github.com/anasqadri-dev/software-quality-engineering/blob/main/LABs/LAB%2309/sqe-gradebook/tests/PercentileTest.java

The JUnit execution results showing all seven tests passed.



THE END